

EXP NO:

DATE:

5.Implement the Digital Signature Algorithm (DSA).

AIM:

ALGORITHM:

231401043

PROGRAM:

```
import hashlib

p = 23

q = 11

g = 4

x = 3

y = pow(g, x, p)

k = 7

msg = input("Enter Message: ")

h = int(hashlib.sha256(msg.encode()).hexdigest(), 16) % q

r = pow(g, k, p) % q

k_inv = pow(k, -1, q)

s = (k_inv * (h + x * r)) % q

print("\nDigital Signature")

print("r =", r)

print("s =", s)

w = pow(s, -1, q)

u1 = (h * w) % q

u2 = (r * w) % q

v = ((pow(g, u1, p) * pow(y, u2, p)) % p) % q

print("\nVerification")

print("v =", v)
```

if v == r:

print("Signature is Valid")

else:

print("Signature is Invalid")

OUTPUT:

```
Enter Message: hello  
Signature = (8, 4)  
Verification Value = 8  
Signature Valid
```

RESULT: